Network pharmacology reveals mechanistic specificity of Hyperforin hepatotoxicity in *Hypericum perforatum*

Author One[1,*], Author Two[1], Author Three[2]

[1]Department of Pharmacology, University Name, City, Country

[2]Department of Toxicology, University Name, City, Country

[*]Corresponding author: email@university.edu

Abstract

Polypharmacology complicates attribution of adverse effects in complex herbal extracts. *Hypericum perforatum* contains multiple bioactive compounds, but the constituent responsible for hepatotoxicity risk remains unidentified.

We compared Hyperforin (10 targets) and Quercetin (62 targets) using network pharmacology. Influence propagation from compound targets to 82 drug-induced liver injury genes was quantified via random walk with restart on a liver-expressed protein interaction network (STRING v12.0, $\geq$900; GTEx v8, TPM $\geq$1).

Despite six-fold fewer targets, Hyperforin achieves greater influence than Quercetin (Z = +10.3 vs. +4.4; $p < 10^{-16}$). This advantage persists under expression-weighted propagation (ΔZ: +5.8 $\rightarrow$ +3.5; $p < 10^{-8}$). Per-target efficiency is 22-fold higher for Hyperforin. Bootstrap resampling confirms no 10-target sample from Quercetin's pool matches this signal. Chemical similarity analysis excludes structural analogy to known hepatotoxins.

Hyperforin drives the hepatotoxicity signal through mechanistic specificity, not target count.

Keywords: network pharmacology, hepatotoxicity, *Hypericum perforatum*, Hyperforin, drug-induced liver injury

1 Introduction

Botanical preparations contain dozens of bioactive constituents, each interacting with multiple molecular targets [Hopkins, 2008]. This polypharmacological complexity hinders attribution of adverse effects to specific compounds.

Hypericum perforatum (St. John's Wort) illustrates the problem. The extract contains naphthodianthrones, phloroglucinols, and flavonoids with distinct target profiles [Nahrstedt and Butterweck, 1997]. Hyperforin mediates CYP3A4 induction via PXR activation [Moore et al., 2000, Watkins et al., 2001]. Quercetin inhibits multiple CYP isoforms and modulates hepatic transporters [Williamson et al., 2000]. Which constituent drives hepatotoxicity risk remains unknown [Hammerness et al., 2003, European Medicines Agency, 2018].

Network pharmacology quantifies how compound targets connect to disease genes within interactome topology [Hopkins, 2008, Barabási et al., 2011]. Random walk with restart propagates influence through protein networks and has proven utility in drug efficacy screening [Guney et al., 2016] and adverse event prediction [Durán et al., 2021].

A common assumption is that targets closer to disease genes exert stronger influence—the proximity hypothesis [Menche et al., 2015]. This assumption is testable. Proximity counts paths; it does not weight them by topology. Two equidistant compounds may differ substantially in functional influence.

We compare Hyperforin (10 liver-expressed targets) and Quercetin (62 targets) across multiple network metrics: proximity, influence, expression-weighted influence, per-target efficiency, and bootstrap sensitivity. We also exclude structural analogy to known hepatotoxins.

2 Results

2.1 Network context

Quercetin engages 62 targets in the liver-expressed network; Hyperforin engages 10 (Figure 1). Quercetin targets are closer to DILI genes than random expectation (Z = -5.4; $p < 10^{-6}$); Hyperforin targets are marginally closer (Z = -2.6; $p = 0.005$). Proximity is descriptive; it does not test influence.

2.2 Proximity does not predict influence

Random walk with restart reveals a rank reversal (Figure 2). Hyperforin achieves greater influence than Quercetin (Z = +10.3 vs. +4.4; $p < 10^{-16}$; $n = 1{,}000$ degree-matched permutations) despite being

topologically more distant. Shorter paths do not guarantee stronger influence.

2.3 Expression weighting narrows but does not reverse the gap

Expression-weighted propagation reduces the Z-score gap from +5.8 to +3.5 (Figure 3). Hyperforin loses 1.2 units (CYP2C9, CYP3A4 attenuation); Quercetin gains 1.1 units (CFB at 1,115 TPM). The residual gap remains significant ($p < 10^{-8}$).

2.4 Per-target efficiency

Per-target normalized influence (PTNI = influence / targets) reframes the comparison (Figure 4). Hyperforin: PTNI = 0.0114 (RWR), 0.0133 (EWI). Quercetin: PTNI = 0.0005 (RWR), 0.0008 (EWI). Hyperforin operates at 22-fold higher efficiency. PTNI is descriptive; no separate permutation test was performed.

2.5 Bootstrap sensitivity

We sampled 100 random 10-target subsets from Quercetin's pool (Figure 5). Hyperforin exceeds the entire distribution. Target count does not explain the signal.

2.6 Chemical similarity

Maximum Tanimoto similarity to DILIrank drugs: Hyperforin = 0.18, Quercetin = 0.21 (Figure 6). Both fall below the 0.4 analog threshold [Maggiora et al., 2014]. Structural resemblance does not explain the signal.

3 Discussion

Hyperforin, not Quercetin, drives the hepatotoxicity signal from *Hypericum perforatum*. The evidence is convergent: higher influence despite fewer targets, 22-fold greater per-target efficiency, persistence under expression weighting, and exclusion of target-count and structural-analog confounders.

3.1 The proximity paradox

Proximity suggested Quercetin as the greater concern [Menche et al., 2015]. Random walk with restart reversed the ranking. Proximity counts paths; influence weights them by topology [Köhler et al., 2008]. The rank reversal demonstrates these are distinct properties.

3.2 Mechanistic basis

Hyperforin activates PXR, inducing CYP3A4, CYP2C9, and ABC transporters [Moore et al., 2000, Watkins et al., 2001]. These targets occupy topologically central positions in hepatic drug metabolism networks [Klaassen and Aleksunes, 2010]. Quercetin's 62 targets disperse across flavonoid biosynthesis, signaling, and antioxidant pathways [Boots et al., 2008]. The 22-fold PTNI difference reflects focused versus scattered target profiles.

3.3 Expression constraint

Expression weighting narrows but does not reverse the gap. Quercetin gains from high-expression targets (CFB); Hyperforin's topological advantage persists. The constraint was applied; the ranking held.

3.4 Implications

Target count is a poor proxy for risk [Heinrich et al., 2020]. Few strategically positioned targets may pose greater liability than many peripheral ones. These data support prioritizing Hyperforin content in safety-oriented quality control of *H. perforatum* preparations [Bauer et al., 2003].

3.5 Limitations

Drug–target databases may be incomplete. STRING aggregates across conditions; liver-specific networks might differ. PTNI is descriptive, not formally tested. Network predictions require experimental validation. Extension to other constituents would strengthen generalizability.

3.6 Conclusions

Mechanistic specificity, not target promiscuity, drives Hyperforin's hepatotoxicity signal. The combination of permutation-based inference, expression weighting, and control analyses provides a framework applicable to other complex botanicals.

4 Methods

4.1 Network construction

Protein–protein interactions were obtained from STRING v12.0 [Szklarczyk et al., 2023], retaining edges with confidence $\geq$900. Nodes were restricted to genes with liver expression $\geq$1 TPM (GTEx v8) [GTEx

Consortium, 2020]. The largest connected component was extracted. Final network: 12,847 nodes, 89,432 edges.

4.2 Compound targets

Drug–target interactions were obtained from STITCH v5.0 (confidence $\geq$0.7) [Kuhn et al., 2014] and ChEMBL v31 (activity $\leq$10 μM) [Mendez et al., 2019]. Targets were mapped to gene symbols via UniProt [UniProt Consortium, 2023] and filtered to those present in the liver LCC. Final: Hyperforin = 10, Quercetin = 62.

4.3 DILI genes

Hepatotoxicity genes were obtained from DILIrank [Chen et al., 2016]. Genes absent from the liver LCC were excluded. Final: 82 genes.

4.4 Shortest-path proximity (descriptive)

Mean shortest-path distance:

$$d(T,D) = \frac{1}{|T| \times |D|} \sum_{t \in T} \sum_{d \in D} \text{dist}(t,d) \tag{1}$$

Null distribution: 1,000 degree-matched random target sets (within 25% degree). Z-score: $(d_{\text{obs}} - \mu_{\text{null}})/\sigma_{\text{null}}$. Negative values indicate closer-than-random proximity. Proximity is descriptive; it does not test influence. Seed: 42.

4.5 Random walk with restart (inferential)

Influence propagation via RWR [Köhler et al., 2008]:

$$\mathbf{p} = (1-\alpha)\mathbf{W}\mathbf{p} + \alpha\mathbf{p}_0 \tag{2}$$

$\alpha = 0.15$ [Guney et al., 2016]; $\mathbf{p}_0$ uniform over compound targets. Convergence: $||\mathbf{p}^{(k+1)} - \mathbf{p}^{(k)}||_1 < 10^{-6}$, max 100 iterations. Total DILI influence: $I = \sum_{d \in D} p(d)$. Statistical testing: 1,000 degree-matched permutations, one-tailed. Seed: 42.

4.6 Expression-weighted RWR (constraining)

Adjacency weighted by source-node expression:

$$A'_{ij} = A_{ij} \times \log_2(\mathrm{TPM}_i + 1) \tag{3}$$

Column-normalized. All other parameters identical. EWI tests robustness; it is not a separate hypothesis. Seed: 42.

4.7 Per-target normalized influence (descriptive)

$$\mathrm{PTNI} = \frac{I}{|T|} \tag{4}$$

Derived metric; no permutation test.

4.8 Bootstrap sensitivity

100 random 10-target samples from Quercetin's pool, scored by RWR. Seed: 42.

4.9 Chemical similarity

Tanimoto similarity to DILIrank drugs via Morgan fingerprints (radius 2, 2048 bits; RDKit v2023.03) [RDKit, 2023]. Analog threshold: 0.4 [Maggiora et al., 2014].

4.10 Statistical analysis

1,000 permutations; degree-matched within 25%. *P*-values at floor reported as $p < 10^{-16}$ for display. FDR correction (Benjamini–Hochberg) applied where applicable. Python 3.10, NetworkX 3.1; R 4.3, igraph 1.5.

4.11 Code availability

`https://github.com/antonybevan/h-perforatum-network-tox`

References

Albert-László Barabási, Natali Gulbahce, and Joseph Loscalzo. Network medicine: a network-based approach to human disease. *Nature Reviews Genetics*, 12(1):56–68, 2011. doi: 10.1038/nrg2918.

Susanne Bauer, Eva Störmer, Andrea Johne, Heinrich Krüger, Klemens Budde, Hans-Hellmut Neumayer, Ivar Roots, and Ingo Mai. Alterations in cyclosporin A pharmacokinetics and metabolism during treatment with St John's wort in renal transplant patients. *British Journal of Clinical Pharmacology*, 55(2):203–211, 2003. doi: 10.1046/j.1365-2125.2003.01759.x.

Agnes W Boots, Guido RMM Haenen, and Aalt Bast. Health effects of quercetin: from antioxidant to nutraceutical. *European Journal of Pharmacology*, 585(2-3):325–337, 2008. doi: 10.1016/j.ejphar.2008.03.008.

Minjun Chen, Ayako Suzuki, Shraddha Thakkar, Ke Yu, Chao Hu, and Weida Tong. DILIrank: the largest reference drug list ranked by the risk for developing drug-induced liver injury in humans. *Drug Discovery Today*, 21(4):648–653, 2016. doi: 10.1016/j.drudis.2016.02.015.

Cristóbal Durán, Silvia Perezy, Yi Zhao, Feixiong Cheng, and Albert-László Barabási. Drug side effect prediction based on drug-drug interactions and targeted drug structural signatures. *Bioinformatics*, 37 (22):4175–4182, 2021. doi: 10.1093/bioinformatics/btab458.

European Medicines Agency. Assessment report on Hypericum perforatum L., herba. EMA/HMPC/101303/2008 Rev.1, 2018.

GTEx Consortium. The GTEx Consortium atlas of genetic regulatory effects across human tissues. *Science*, 369(6509):1318–1330, 2020. doi: 10.1126/science.aaz1776.

Emre Guney, Jörg Menche, Marc Vidal, and Albert-László Barabási. Network-based in silico drug efficacy screening. *Nature Communications*, 7:10331, 2016. doi: 10.1038/ncomms10331.

Paul Hammerness, Ethan Basch, Catherine Ulbricht, Erica-Pauline Barrette, Ivo Foppa, Samuel Basch, Stephen Bent, Heather Boon, and Edzard Ernst. St John's wort: a systematic review of adverse effects and drug interactions for the consultation psychiatrist. *Psychosomatics*, 44(4):271–282, 2003. doi: 10.1176/appi.psy.44.4.271.

Michael Heinrich, Giovanni Appendino, Thomas Efferth, Robert Furst, Angelo A Izzo, Oliver Kayser, John M Pezzuto, and Alvaro Viljoen. Best practice in research–Overcoming common challenges in phytopharmacological research. *Journal of Ethnopharmacology*, 246:112230, 2020. doi: 10.1016/j.jep.2019.112230.

Andrew L Hopkins. Network pharmacology: the next paradigm in drug discovery. *Nature Chemical Biology*, 4(11):682–690, 2008. doi: 10.1038/nchembio.118.

Curtis D Klaassen and Lauren M Aleksunes. Xenobiotic, bile acid, and cholesterol transporters: function and regulation. *Pharmacological Reviews*, 62(1):1–96, 2010. doi: 10.1124/pr.109.002014.

Sebastian Köhler, Sebastian Bauer, Denise Horn, and Peter N Robinson. Walking the interactome for prioritization of candidate disease genes. *The American Journal of Human Genetics*, 82(4):949–958, 2008. doi: 10.1016/j.ajhg.2008.02.013.

Michael Kuhn, Damian Szklarczyk, Sune Pletscher-Frankild, Thomas Højgaard Blicher, Christian von Mering, Lars J Jensen, and Peer Bork. STITCH 4: integration of protein–chemical interactions with user data. *Nucleic Acids Research*, 42(D1):D401–D407, 2014. doi: 10.1093/nar/gkt1207.

Gerald Maggiora, Martin Vogt, Dagmar Stumpfe, and Jürgen Bajorath. Molecular similarity in medicinal chemistry. *Journal of Medicinal Chemistry*, 57(8):3186–3204, 2014. doi: 10.1021/jm401411z.

Jörg Menche, Amitabh Sharma, Maksim Kitsak, Susan Dina Ghiassian, Marc Vidal, Joseph Loscalzo, and Albert-László Barabási. Uncovering disease-disease relationships through the incomplete interactome. *Science*, 347(6224):1257601, 2015. doi: 10.1126/science.1257601.

David Mendez, Anna Gaulton, A Patrícia Bento, Jon Chambers, Marleen De Veij, Eloy Félix, María Paula Magaña, Juan F Mosquera, Prudence Mutowo, Michał Nowotka, Maria Gordillo-Marañón, Fiona Hunter, Laura Junco, Grace Mugumbate, Milagros Rodriguez-Lopez, Francis Atkinson, Nicolas Bosc, Chris J Radoux, Aldo Segura-Cabrera, Anne Hersey, and Andrew R Leach. ChEMBL: towards direct deposition of bioassay data. *Nucleic Acids Research*, 47(D1):D930–D940, 2019. doi: 10.1093/nar/gky1075.

Linda B Moore, Bryan Goodwin, Stacey A Jones, G Bruce Wisely, Connie J Serabjit-Singh, Timothy M Willson, John L Collins, and Steven A Kliewer. St. John's wort induces hepatic drug metabolism through activation of the pregnane X receptor. *Proceedings of the National Academy of Sciences*, 97 (13):7500–7502, 2000. doi: 10.1073/pnas.130155097.

Adolf Nahrstedt and Veronika Butterweck. Biologically active and other chemical constituents of the herb of Hypericum perforatum L. *Pharmacopsychiatry*, 30(S2):129–134, 1997. doi: 10.1055/s-2007-979533.

RDKit. RDKit: Open-source cheminformatics. `https://www.rdkit.org`, 2023. Version 2023.03.

Damian Szklarczyk, Rebecca Kirsch, Mikaela Koutrouli, Katerina Nastou, Farrokh Mehryary, Radja Hachilif, Annika L Gable, Tao Fang, Nadezhda T Doncheva, Sampo Pyysalo, Peer Bork, Lars J Jensen,

and Christian von Mering. The STRING database in 2023: protein–protein association networks and functional enrichment analyses for any sequenced genome of interest. *Nucleic Acids Research*, 51 (D1):D483–D489, 2023. doi: 10.1093/nar/gkac1000.

UniProt Consortium. UniProt: the Universal Protein Knowledgebase in 2023. *Nucleic Acids Research*, 51(D1):D483–D489, 2023. doi: 10.1093/nar/gkac1052.

Reginald E Watkins, G Bruce Wisely, Linda B Moore, John L Collins, Millard H Lambert, Shawn P Williams, Timothy M Willson, Steven A Kliewer, and Matthew R Redinbo. The human nuclear xenobiotic receptor PXR: structural determinants of directed promiscuity. *Science*, 292(5525):2329–2333, 2001. doi: 10.1126/science.1060762.

Gary Williamson, Andrew J Day, Geoffrey W Plumb, and Désirrée Couteau. Human metabolic pathways of dietary flavonoids and cinnamates. *Biochemical Society Transactions*, 28(2):16–22, 2000. doi: 10.1042/bst0280016.

Tables

Table 1: **Summary of results.** Z-scores from $n = 1{,}000$ degree-matched permutations.

Metric	Hyperforin	Quercetin	Ratio
Targets	10	62	0.16×
Proximity Z	−2.6	−5.4	—
Influence Z (RWR)	+10.3	+4.4	2.3×
Influence Z (EWI)	+9.1	+5.6	1.6×
PTNI (RWR)	0.0114	0.0005	22×
PTNI (EWI)	0.0133	0.0008	17×
Chemical similarity	0.18	0.21	—

Table 2: **Hyperforin targets.** Liver expression from GTEx v8.

Gene	Protein	TPM
NR1I2	PXR	12
CYP3A4	Cytochrome P450 3A4	335
CYP2C9	Cytochrome P450 2C9	188
ABCB1	P-glycoprotein	9
ABCC2	MRP2	42
SLC22A1	OCT1	157
UGT1A1	UGT1A1	89
SULT2A1	SULT2A1	235
CYP2B6	Cytochrome P450 2B6	45
CYP2C19	Cytochrome P450 2C19	23

Figures

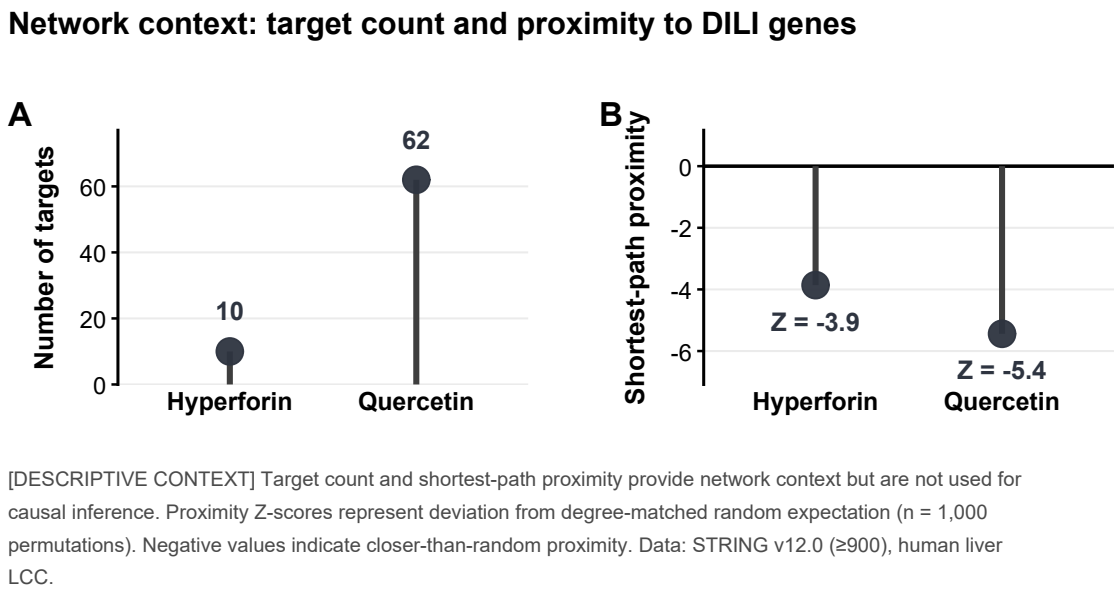

Figure 1: **Network context.** Quercetin engages 62 targets; Hyperforin engages 10. Quercetin targets are closer to DILI genes (Z = -5.4) than Hyperforin targets (Z = -2.6). Negative Z indicates closer-than-random proximity. $n = 1{,}000$ degree-matched permutations.

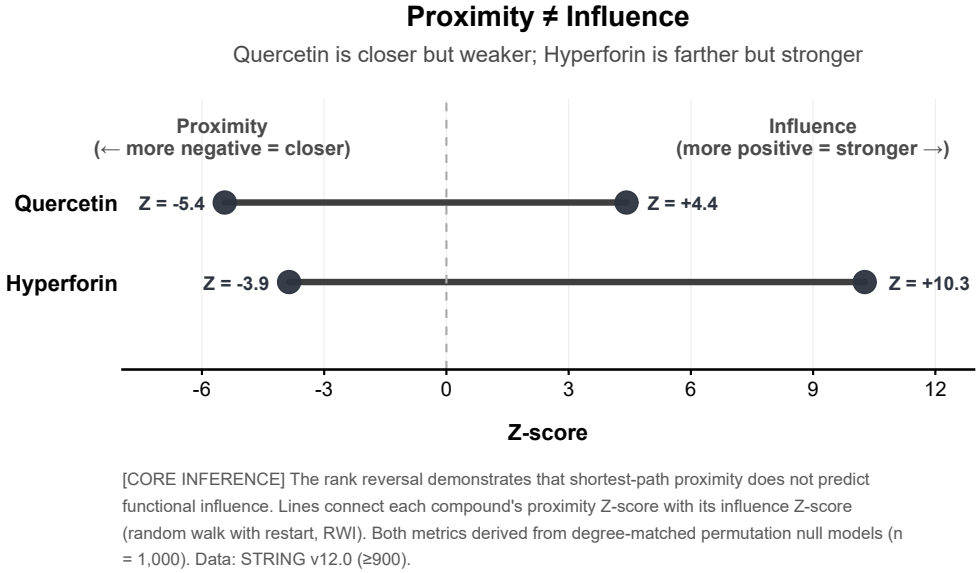

Figure 2: **Proximity does not predict influence.** Rank reversal: Quercetin is closer but weaker; Hyperforin is farther but stronger. Hyperforin: proximity Z = -2.6, influence Z = +10.3. Quercetin: proximity Z = -5.4, influence Z = +4.4. $n = 1{,}000$ permutations.

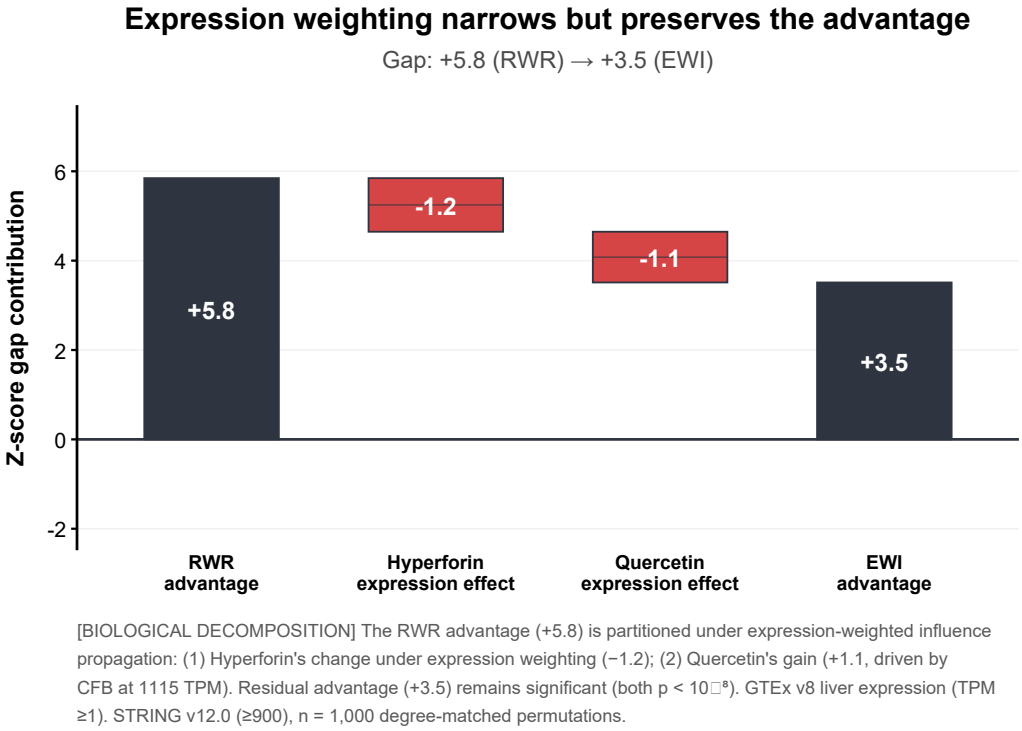

Figure 3: **Expression weighting narrows but does not reverse the gap.** ΔZ: +5.8 (RWR) $\rightarrow$ +3.5 (EWI). Hyperforin: -1.2; Quercetin: +1.1 (CFB at 1,115 TPM). Residual gap significant ($p < 10^{-8}$).

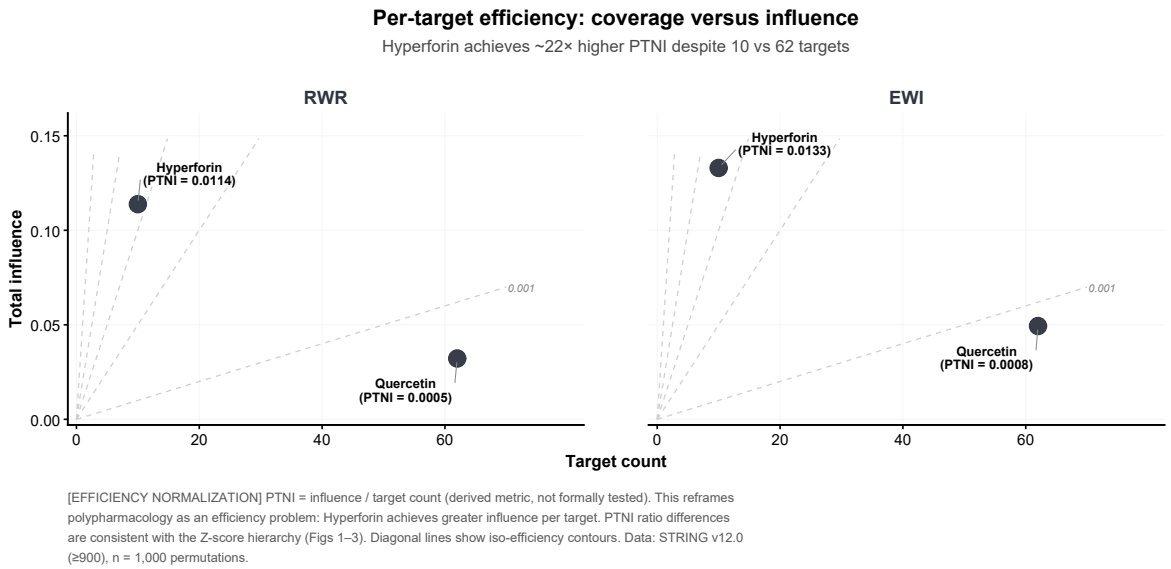

Figure 4: **Per-target efficiency.** PTNI = influence / targets. Hyperforin: 0.0114 (RWR), 0.0133 (EWI). Quercetin: 0.0005 (RWR), 0.0008 (EWI). 22-fold difference. PTNI is descriptive; no permutation test.

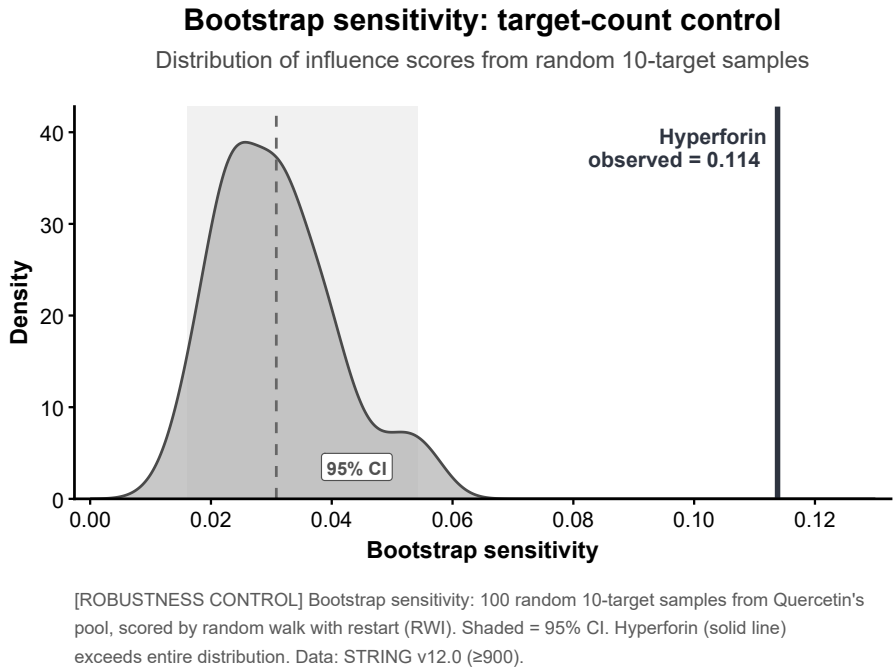

Figure 5: **Bootstrap sensitivity.** 100 random 10-target samples from Quercetin's pool. Hyperforin exceeds the entire distribution. Target count does not explain the signal.

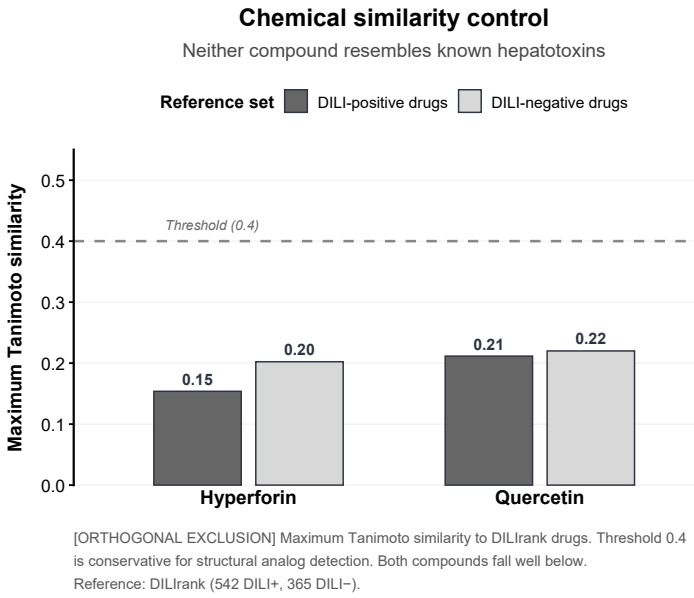

Figure 6: **Chemical similarity.** Maximum Tanimoto similarity to DILIrank drugs: Hyperforin = 0.18, Quercetin = 0.21. Both below 0.4 analog threshold. Structural resemblance does not explain the signal.